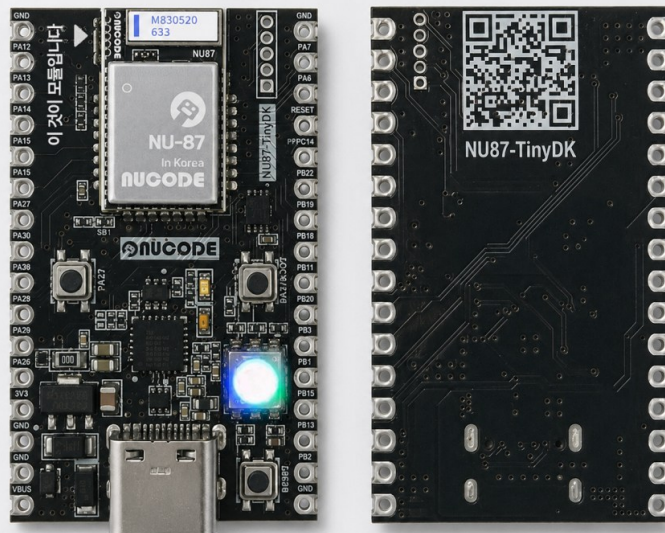


NU-87 TinyDK Start Guide



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1. OVERVIEW

This guide walks you through the basic setup required to use the NU-87 TinyDK with the Arduino IDE, and how to confirm that your board is working correctly. It is intended to take you from a freshly unboxed board to a verified development environment.

The guide covers:

- Installing the Arduino IDE Board Package
- Selecting the NU-87 TinyDK board
- Configuring the upload mode
- Enabling 5 ghz Wi-Fi
- Running a basic RGB LED blink test

The NU-87 TinyDK schematic and NU-87 module datasheet are available at the links below:

[NU-87 TinyDK Schematic](#)

[NU-87 Module Datasheet](#)

2. REQUIREMENTS

You will need the following to test the NU-87 TinyDK:

- NU-87 TinyDK
- USB Cable
- Arduino IDE
- PC

* Note: Charge-only USB cables can prevent the board from being detected or cause uploads to fail, so be sure to use a cable that supports data communication.

3. ARDUINO IDE SETUP

To use the NU-87 TinyDK with the Arduino IDE, you first need to install the Realtek Ameba board package.

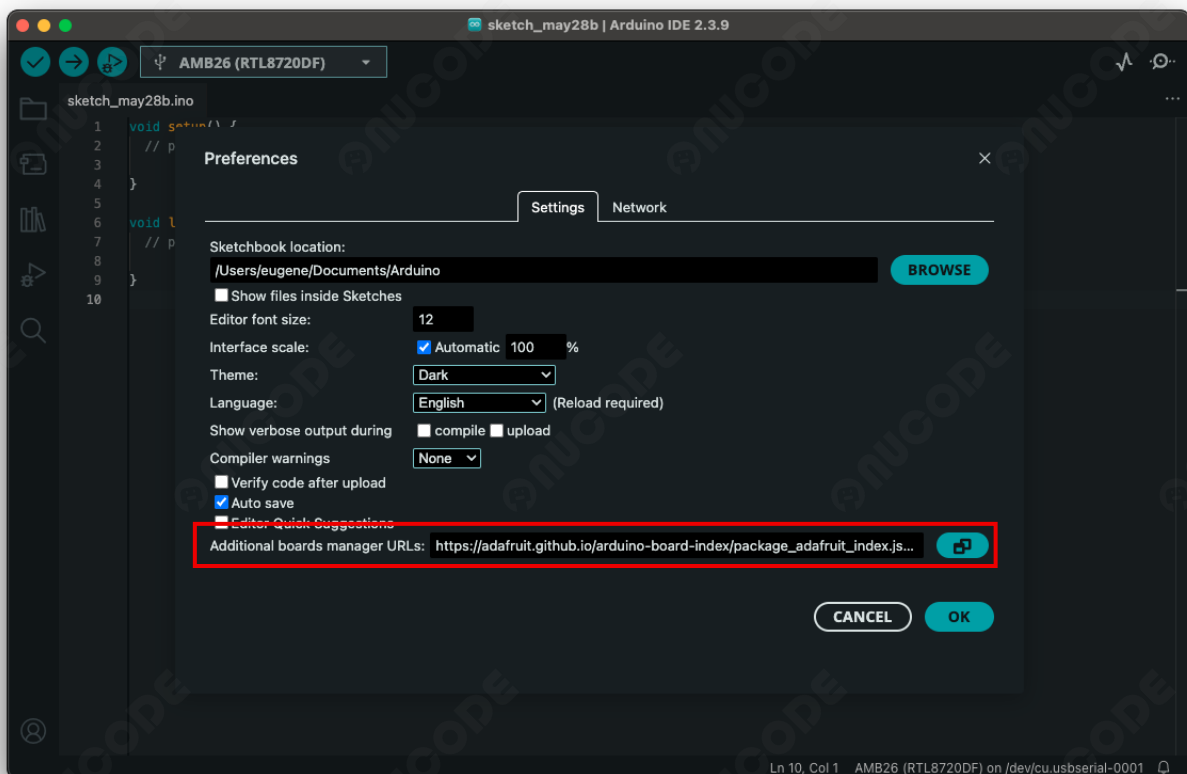
3.1. ADDING THE BOARDS MANAGER URL

Launch the Arduino IDE and go to:

- Windows / Linux: File > Preferences
- macOS: Arduino IDE > Preferences

Add the following URL under Additional Boards Manager URLs:

https://github.com/Ameba-AIoT/ameba-arduino-d/raw/master/Arduino_package/package_realtek_amebad_index.json

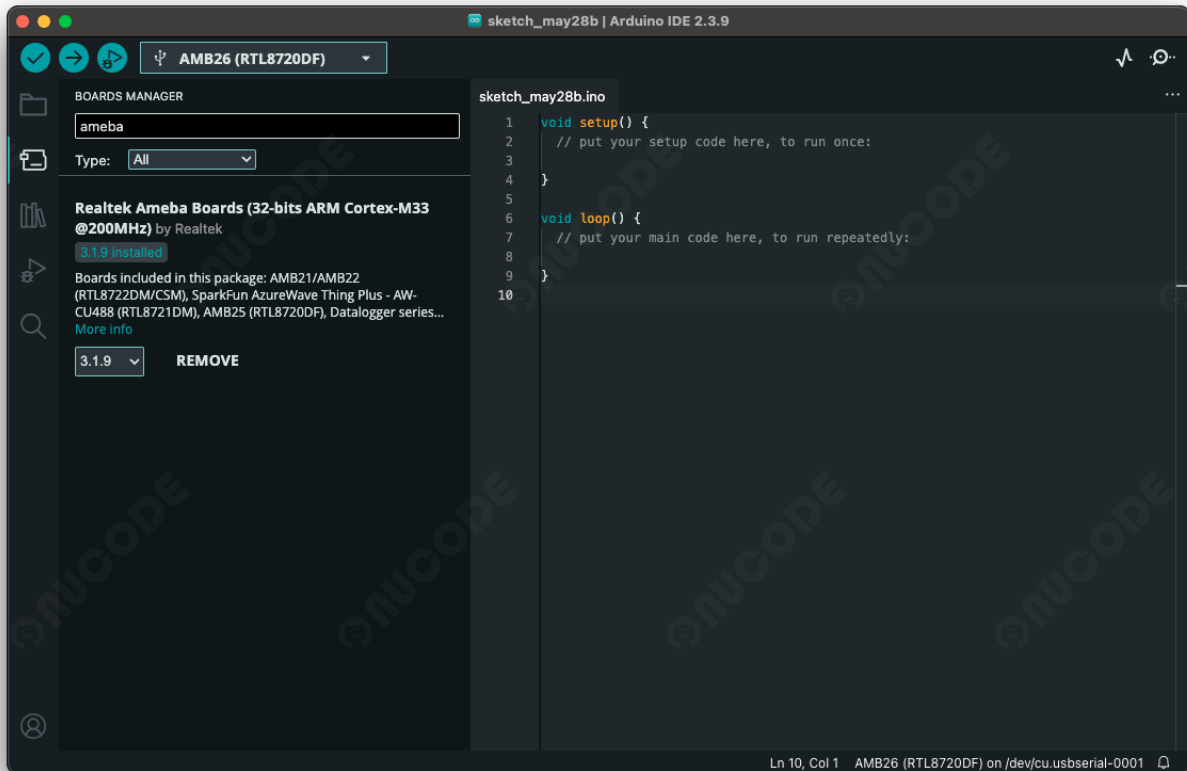


3.2. INSTALLING THE BOARD PACKAGE

Go to Tools > Board > Boards Manager.

In the search box, enter 'Realtek Ameba Boards', then install the following package from the results:

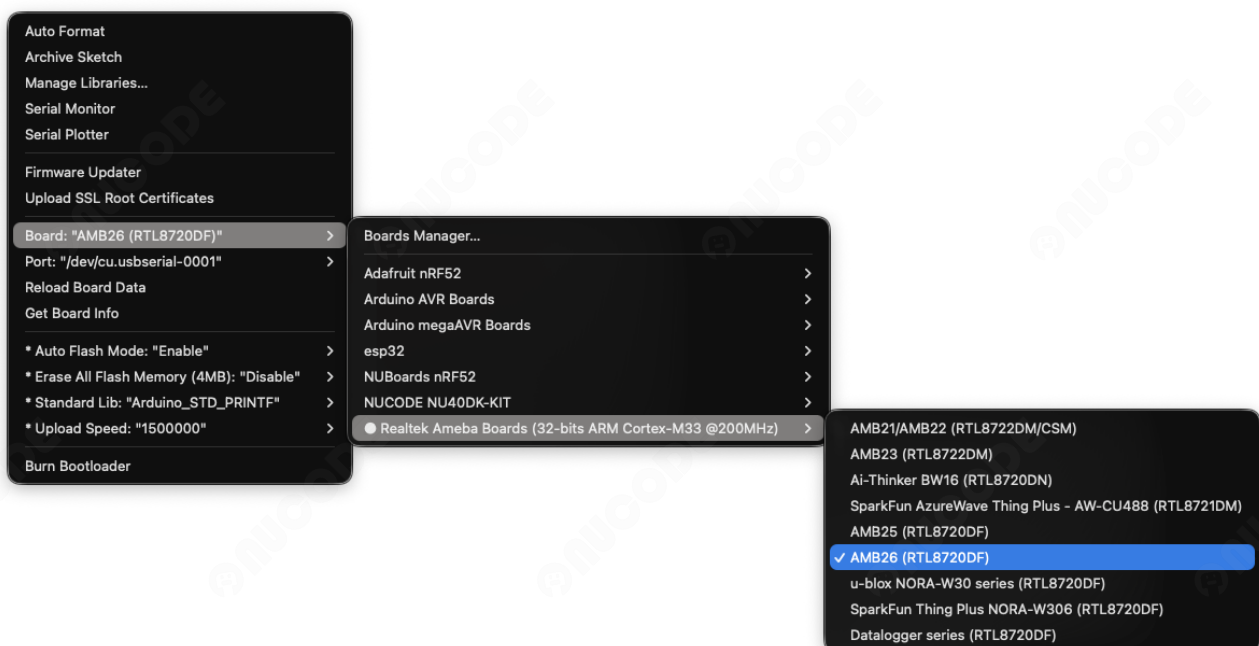
Realtek Ameba Boards (32-bits ARM Cortex-M33 @200MHz) by Realtek



3.3. SELECTING THE BOARD

Once the board package is installed, select the board in the Arduino IDE:

Tools > Board > Realtek Ameba Boards > AMB26 (RTL8720DF)



3.4. SELECTING THE PORT

Connect the NU-87 TinyDK to your PC, then select the port under Tools > Port.



The port name varies depending on your environment:

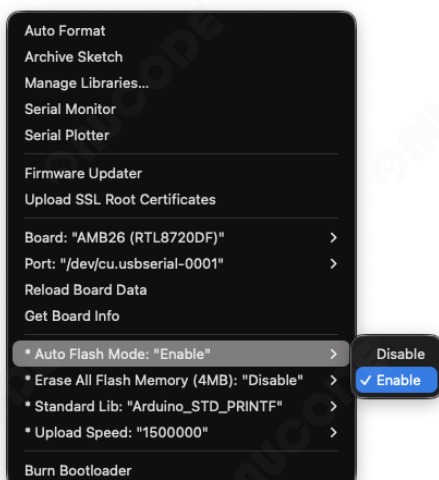
- Windows: COMx
- macOS: /dev/cu.usbserial-xxxx or /dev/cu.usbmodemxxxx
- Linux: /dev/ttyUSBx or /dev/ttyACMx

4. AUTO FLASH MODE

The NU-87 TinyDK supports an Auto Flash Mode setting that depends on how you want to upload.

4.1. AUTO FLASH MODE: ENABLE

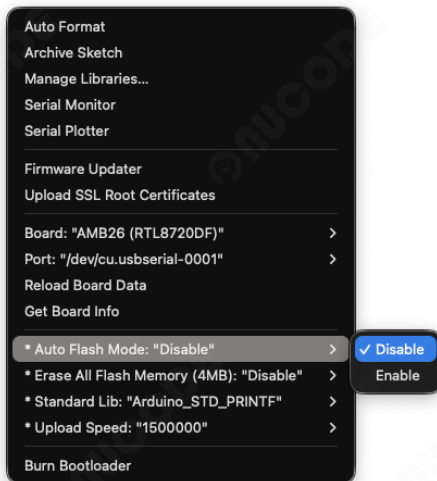
Select Tools > Auto Flash Mode > Enable.



When Auto Flash Mode is set to Enable, the board automatically enters flash mode during upload. Enable is recommended for typical testing.

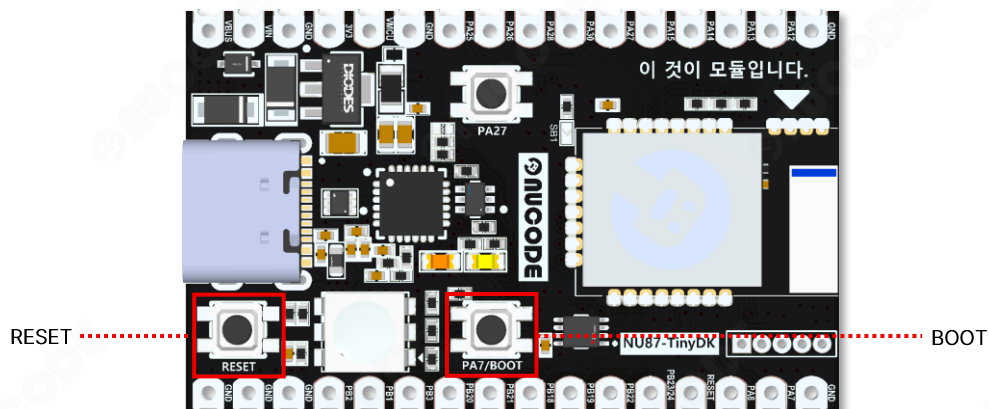
4.2. AUTO FLASH MODE: DISABLE

Select Tools > Auto Flash Mode > Disable.



When Auto Flash Mode is Disable, you must enter flash mode manually.

Within about five seconds of starting the upload, hold down the BOOT button, press and release the RESET button, and then release the BOOT button.



5. 5GHZ WI-FI ENABLE

To use 5 GHz Wi-Fi on the NU-87 TinyDK, you need to upload a sketch that sets the channel plan.

Paste the following code into the Arduino IDE and upload it:

```
extern "C" {
#include "wifi_conf.h"
}

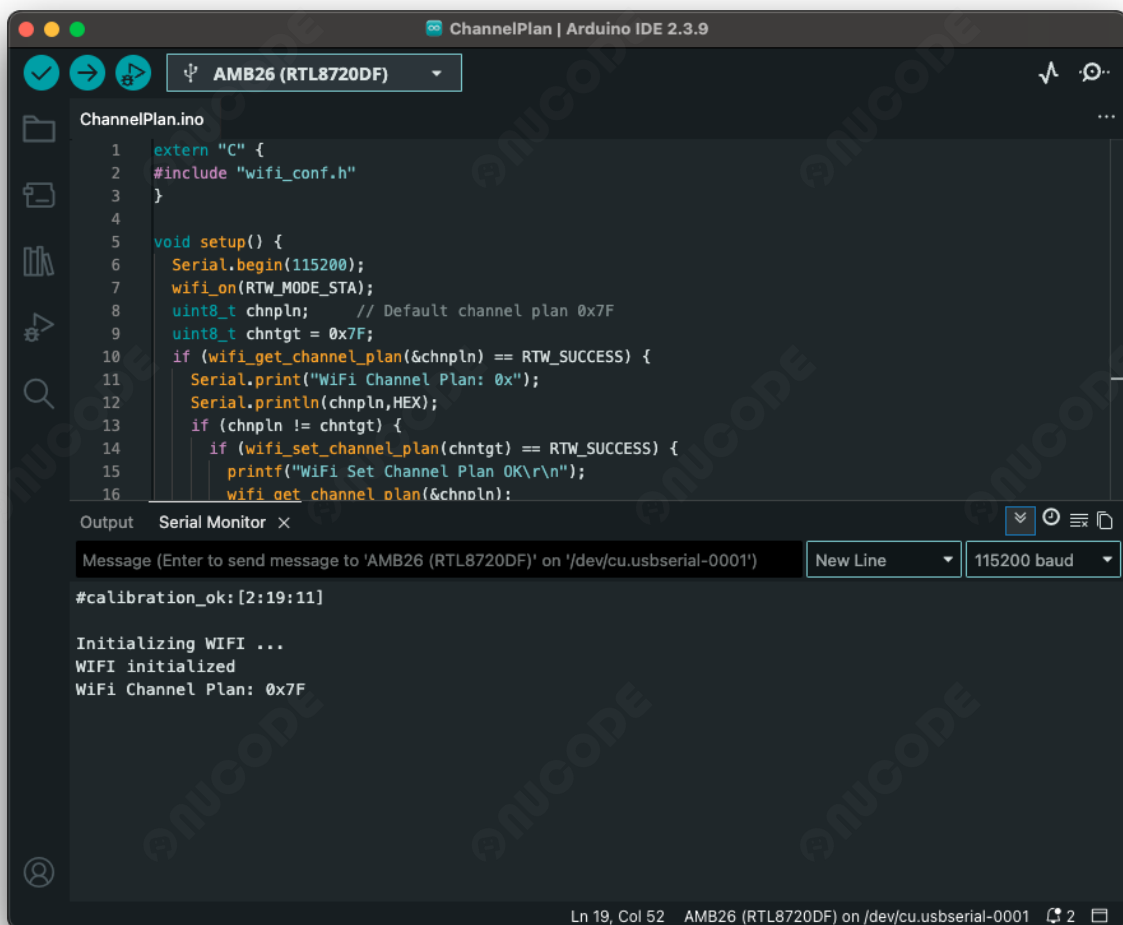
void setup() {
  Serial.begin(115200);
  wifi_on(RTW_MODE_STA);
  uint8_t chnpln; // Default channel plan 0x7F
```

```
uint8_t chntgt = 0x7F;
if (wifi_get_channel_plan(&chnpln) == RTW_SUCCESS) {
    Serial.print("WiFi Channel Plan: 0x");
    Serial.println(chnpln, HEX);
    if (chnpln != chntgt) {
        if (wifi_set_channel_plan(chntgt) == RTW_SUCCESS) {
            printf("WiFi Set Channel Plan OK\r\n");
            wifi_get_channel_plan(&chnpln);
            printf("WiFi Channel Plan: 0x%x\r\n", chnpln);
        } else {
            printf("WiFi Set Channel Plan Failed\r\n");
        }
    }
}

void loop() {
```

After uploading, open the Serial Monitor and set the baud rate to 115200.

When the channel plan is set correctly, the Serial Monitor displays output as following:



6. RGB LED BLINK TEST

To confirm that the board is working, upload the RGB LED blink example.
Paste the following code into the Arduino IDE and upload it:

```
void setup() {  
  pinMode(LED_R, OUTPUT);  
  pinMode(LED_G, OUTPUT);  
  pinMode(LED_B, OUTPUT);  
}  
  
void blink() {  
  digitalWrite(LED_R, HIGH);  
  delay(1000);  
  digitalWrite(LED_R, LOW);  
  delay(1000);  
  
  digitalWrite(LED_G, HIGH);  
  delay(1000);  
  digitalWrite(LED_G, LOW);  
  delay(1000);  
  
  digitalWrite(LED_B, HIGH);  
  delay(1000);  
  digitalWrite(LED_B, LOW);  
  delay(1000);  
}  
  
void loop() {  
  blink();  
  delay(1000);  
}
```

When everything is working, the RGB LED cycles in the order RED → GREEN → BLUE, with each color lit for about one second before turning off.

7. TROUBLESHOOTING

7.1. THE BOARD PACKAGE DOES NOT APPEAR IN SEARCH

Make sure the URL is entered exactly under **Additional Boards Manager URLs**.

If the Boards Manager **Type** filter is set to **Installed**, only packages you have already installed will be shown. If you can't find the package, change **Type** to **All** and search again.

7.2. UPLOAD FAILS

Check the following in order:

- confirm that the USB cable supports data transfer
- confirm that the correct port is selected
- confirm that the board is set to AMB26 (RTL8720DF)
- confirm that Auto Flash Mode is set to Enable

If Auto Flash Mode is Disable, enter flash mode manually using the BOOT + RESET sequence.

7.3. BW16 EXAMPLE DOES NOT WORK

Any Arduino IDE example with "BW16" in its name is specific to the Ai-Thinker BW16. Because the NU-87 TinyDK is used as the **AMB26 (RTL8720DF)** board, running a BW16-only example may produce the following message:

Error. This example is only for board BW16

When testing the NU-87 TinyDK, use the RGB LED blink code provided in this guide instead.

RELATED DOCUMENTATION

NUCODE Documentation archive: <https://nuworks.io/wiki/software>

REVISION HISTORY

Date	Version	Release Notes
2026-05-28	V1.0	First release

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